

Mathematics Class 11

Federal Board, 2026 Solved

Math Paper of 2nd Annual

Exam, 2025 Live Session

04 | 7 Days Left

Coming 3 Days Plan In Sha' ALLAH!

- ▶ On Friday, We Will Do Chapter # 04 & 06 Most Important Short and Long Questions in a YouTube LIVE STREAM at 7:30 PM onwards
- ▶ On Saturday, We will Solve Most Important Short and Long Questions of Chapter # 03, 07 & 08 & 09 in a Marathon YouTube LIVE Stream starting from 9:30 AM Onwards
- ▶ On Sunday, We will Solve All Exercises MCQs in a Marathon YouTube Live Session starting from 9:30 AM onwards

	Question	A	B	<u>C</u>	D
✓ 1.	The factorization of $z^2 + a^2$ over complex numbers is:	$(z + a)(z - a)$	$z(z + a^2)$	$\frac{(z + ia)}{(z - ia)}$	$(z - ia)^2$
✓ 2.	If $z = x + iy$, then in polar form $z = r(\cos\theta + i \sin\theta)$, where r equals:	$x + y$	$x^2 + y^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}\left(\frac{y}{x}\right)$
3.	Which of the following is <u>NOT</u> an elementary row operation?	Swapping two rows	Multiplying a row by a nonzero scalar	Adding a multiple of one row to another	Raising a row to a power
4.	The system $x + y + z = 2, x + y + z = 3, x + y + z = 4$ is:	Dependent	Inconsistent	Consistent with infinitely many solutions	Consistent with a unique solution
5.	In an arithmetic sequence, if $a_1 = 3$ and $d = 2$, then 5th term is:	9	11	12	13

$$z^2 + a^2 = z^2 - (-a^2)$$

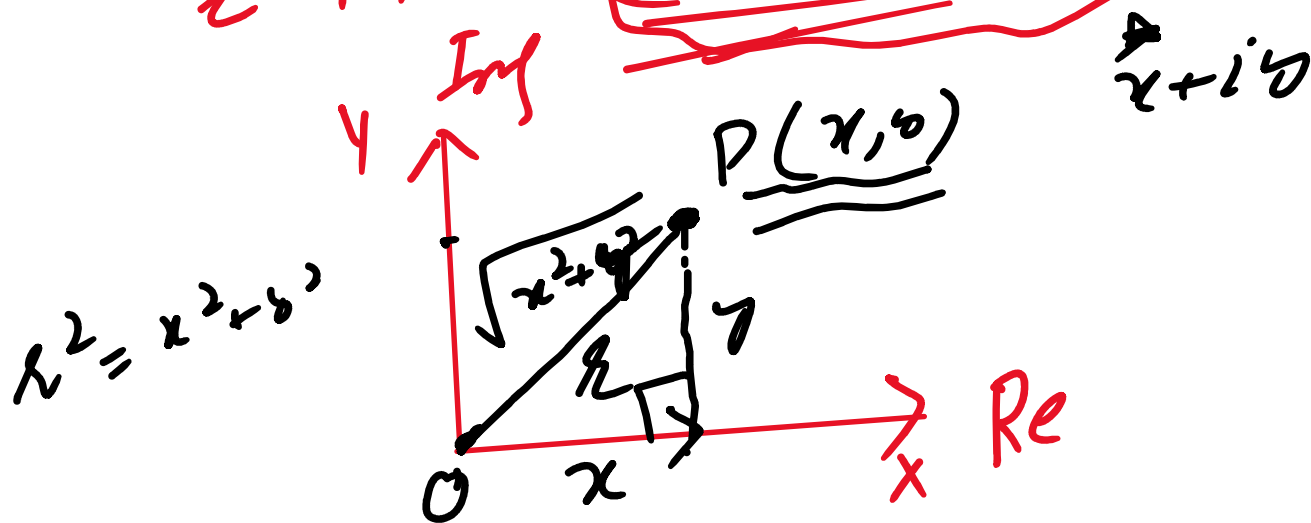
- 2 1, -2, 2, 1

$$i^2 = -1$$

$$z + a = z - (-a)$$

$$(z + ai)(z - ai) = z^2 - (ai)^2$$

$$z^2 + 9 = (z + 3i)(z - 3i)$$



$$r = |z| = \sqrt{x^2 + y^2}$$

4. The system $x + y + z = 2, x + y + z = 3, x + y + z = 4$ is:	Dependent	Inconsistent	Consistent with infinitely many solutions	Consistent with a unique solution
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$$\begin{cases} x + y + z = 2 \\ x + y + z = 3 \\ 0 = -1 \end{cases}$$

No solution

Consistent

unique

infinite

unique

infinite

$$x + y + z = 3$$

$$2y + z = 4$$

$$0z = 2$$

$$z = 0, 1, 2, \dots$$

Infinite

$$0z = 0 \checkmark \text{ infinite many}$$

$$z = t$$

5. In an arithmetic sequence, if $a_1 = 3$ and $d = 2$, then 5th term is:

9	11	12	13
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Formula Sheet Chapter #1

$$a_1 = 3 \quad d = 2$$

$$a_5 = ?$$

3, 5, 7, 9, 11, ...

6, 7, 4

3, 0, +, 1, 1, 1

7 8 4

$$a_n = a_1 + (n-1)d$$

$$a_5 = 3 + (5-1)(2) = 3 + 8 \in \mathbb{N}$$

H.P 1, 2, 3, 4
Rec

6. Which of the following sequences is Harmonic?	2, 4, 6, 8, ...	3, 6, 12, 24, ...	$\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$	1, 3, 5, 7, ...
7. The coefficient of x^2 in $(1 + 2x)^3$ is:	12	18	24	30
8. According to the principle of mathematical induction, the base step is to verify for:	$n = 1$	$n = 2$	$n = k$	$n = k + 1$
9. If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, then $z_1 z_2$ equals:	$r_1 r_2 e^{i(\theta_1 + \theta_2)}$	$r_1 r_2 e^{i(\theta_1 - \theta_2)}$	$\frac{r_1 r_2}{e^{i(\theta_1 + \theta_2)}}$	$\frac{r_1 r_2}{e^{i(\theta_1 - \theta_2)}}$
10. The last digit of 7^{10} is:	1	3	7	9

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

7. The coefficient of x^2 in $(1 + 2x)^3$ is:

12

18

24

30

$$\binom{3}{0} + \binom{3}{1}(2x) + \binom{3}{2}(2x)^2 + \binom{3}{3}(2x)^3$$

$$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} a^0 b^n$$

$$\binom{n}{1} = \binom{n}{n-1} \quad \checkmark \quad \binom{10}{4} = \binom{10}{6}$$

Base-step set $n=1 \quad \checkmark$

Inductive step $\rightarrow$ P(k) is true &
 $\implies P(k+1)$ must also be true

9. If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, then $z_1 z_2$ equals:	$r_1 r_2 e^{i(\theta_1 + \theta_2)}$	$r_1 r_2 e^{i(\theta_1 - \theta_2)}$	$\frac{r_1 r_2}{e^{i(\theta_1 + \theta_2)}}$	$\frac{r_1 r_2}{e^{i(\theta_1 - \theta_2)}}$
10. The last digit of 7^{10} is:	1	3	7	9

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$\downarrow$ Real $\downarrow$ Imag

Eye for
Euler

$$r_1 r_2 e^{i\theta_1} \cdot e^{i\theta_2}$$

~~$x \cdot x = x^2$~~

$$\underbrace{x \cdot x}_x = x^2$$

~~$x + y = xy$~~

$$x \cdot x = x^2$$

$$\underbrace{x \cdot x}_u = x^u$$

$$x \cdot x = x^2$$

9. If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, then $z_1 z_2$ equals:	$r_1 r_2 e^{i(\theta_1 + \theta_2)}$	$r_1 r_2 e^{i(\theta_1 - \theta_2)}$	$\frac{r_1 r_2}{e^{i(\theta_1 + \theta_2)}}$	$\frac{r_1 r_2}{e^{i(\theta_1 - \theta_2)}}$
10. The last digit of 7^{10} is:	1	3	7	9

Handwritten work for problem 10:

7^{103}
 $7^1, 7^2, 7^3, 7^4, 7^5, \dots$
 $7^1 = 7$
 $7^2 = 49$
 $7^3 = 343$
 $7^4 = 2401$
 $7^5 = 16807$
 $7^6 = 117649$
 $7^7 = 823543$
 $7^8 = 5764801$
 $7^9 = 40353607$
 $7^{10} = 282475249$
 The last digit is 9.

Another method:

$$7^{103} = 7^{100} \cdot 7^3$$

$$7^{100} = (7^4)^{25} = 2401^{25}$$

$$2401 \equiv 1 \pmod{10}$$

$$2401^{25} \equiv 1^{25} \equiv 1 \pmod{10}$$

$$7^{103} \equiv 1 \cdot 343 \equiv 3 \pmod{10}$$
 Wait, this contradicts the previous result. Let's recheck the pattern:

$$7^1 \equiv 7$$

$$7^2 \equiv 9$$

$$7^3 \equiv 3$$

$$7^4 \equiv 1$$

$$7^5 \equiv 7$$

$$7^6 \equiv 9$$

$$7^7 \equiv 3$$

$$7^8 \equiv 1$$

$$7^9 \equiv 7$$

$$7^{10} \equiv 9$$

 The cycle is 4. $103 \div 4 = 25$ remainder 3. So the last digit is the same as 7^3 , which is 3.

Another method:

$$7^{103} = 7^{100} \cdot 7^3$$

$$7^{100} = (7^4)^{25} = 2401^{25}$$

$$2401 \equiv 1 \pmod{10}$$

$$2401^{25} \equiv 1^{25} \equiv 1 \pmod{10}$$

$$7^{103} \equiv 1 \cdot 343 \equiv 3 \pmod{10}$$
 Wait, this contradicts the previous result. Let's recheck the pattern:

$$7^1 \equiv 7$$

$$7^2 \equiv 9$$

$$7^3 \equiv 3$$

$$7^4 \equiv 1$$

$$7^5 \equiv 7$$

$$7^6 \equiv 9$$

$$7^7 \equiv 3$$

$$7^8 \equiv 1$$

$$7^9 \equiv 7$$

$$7^{10} \equiv 9$$

 The cycle is 4. $103 \div 4 = 25$ remainder 3. So the last digit is the same as 7^3 , which is 3.

$$\begin{array}{r} 36 \\ 4 \end{array}$$

$$\begin{array}{r} 8 \downarrow \\ 23 \\ \hline 20 \end{array}$$

Find the unit digit of:

(i) 17^{203}

(ii) 29^{26}

(iii) 36^{307}

$$7^{203}$$



$$\begin{array}{r} 50 \\ 4 \overline{) 203} \\ \underline{200} \\ 3 \end{array}$$

$$\begin{array}{r} 3 \\ 7 \overline{) 21} \\ \underline{14} \\ 7 \end{array}$$

$$25^{26}$$

$$\begin{array}{r} 1 \quad 2 \quad 3 \\ 9 \downarrow \quad 9 \downarrow \quad 9 \downarrow \\ 19 \quad 1 \end{array}$$

$$9^{26} \rightarrow 1$$

$$36^{307}$$

$$6^{307} \rightarrow 6$$

$$\begin{array}{r} 51 \\ 6 \overline{) 307} \\ \underline{306} \\ 1 \end{array}$$

$$\begin{array}{r} 1 \quad 2 \\ 6 \downarrow \quad 6 \downarrow \\ 36 \quad 36 \end{array}$$

65700
 011

11. Remainder of $f(x) = x^3 - 2$ when divided by $(x - 1)$ is	-3	-1	1	2
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$$x - 1 = 0$$

$$x = 1$$

$$1 - 2 = -1$$

12. Which one of the following is a factor of $x^3 - 6x^2 + 11x - 6$?	$x + 1$	$x - 1$	$x - 4$	$x + 3$
13. Which of the following is WRONG ?	$\sin^2 x + \cos^2 x = 1$	$1 - \cos^2 x = \sin^2 x$	$\sin(x + y) = \sin x + \sin y$	$1 + \cot^2 x = \csc^2 x$
14. The maximum value of $3 + 4 \sin \theta$ is:	3	4	7	-1
15. The period of $y = \sin x$ (in radians) is:	$\frac{\pi}{2}$	π	2π	4π
16. $\sin \frac{\pi}{3} \cos \frac{\pi}{6} =$	$\frac{1}{4}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{4}$	$\frac{3}{4}$

$$x = 2$$

$$(x - 2)$$

$$x = 3$$

$$(x - 3)$$

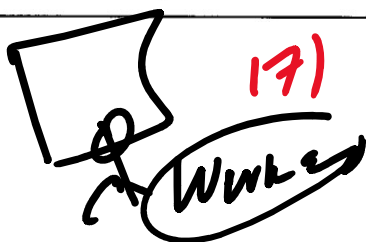
$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin x + \sin y$$

14. The maximum value of $3 + 4 \sin \theta$ is:	3	4	7	-1
15. The period of $y = \sin x$ (in radians) is:	$\frac{\pi}{2}$	π	2π	4π
16. $\sin \frac{\pi}{3} \cos \frac{\pi}{6} =$	$\frac{1}{4}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{4}$	$\frac{3}{4}$

$3 + 4 \sin \theta$ → max val
 $3 + |4|$
 $3 - 2 \sin \left(\frac{3\pi}{4} - \frac{\pi}{3} \right)$
 $\frac{2\pi}{3} = \frac{2\pi \times 4}{3}$
 $\cos 90^\circ = 0$

17. For what value of x , vectors $2\hat{i} - \hat{j} + 3\hat{k}$ and $x\hat{i} + 2\hat{j} - 2\hat{k}$ are orthogonal (mutually perpendicular)?	-2	1	2	4
18. A constant force acts on an object is perpendicular to the displacement, work done is:	Maximum	Zero	Negative	$ \vec{F} \vec{d} $
19. If there is an option for 3 Shirts and 4 Trousers then number of outfits is:	12	10	7	1
20. Number of permutations of 5 distinct objects taken 3 at a time is:	10	20	30	60



$2x - 2 - 6 = 0$
 $2x = 8$

$9.6 = 0$

$$W = \vec{F} \cdot \vec{d} = \underline{\underline{Fdcos\theta}}$$



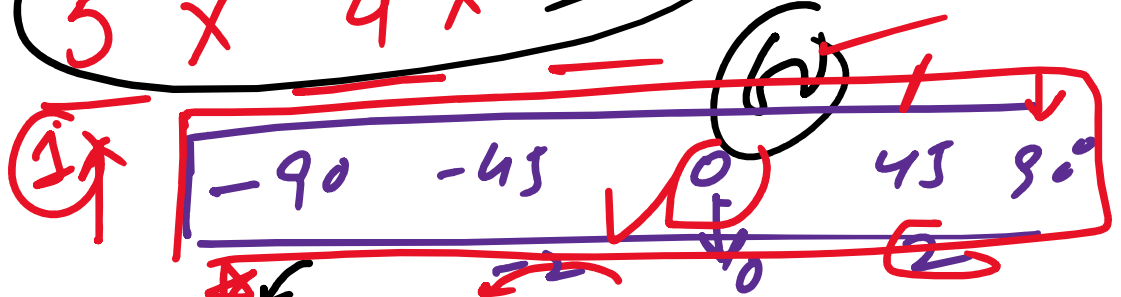
$$3 \times 4 = 12$$



5 numbers



$$5 \times 4 \times 3$$



Draw the graph of $y = 2\sin 2x$; where $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

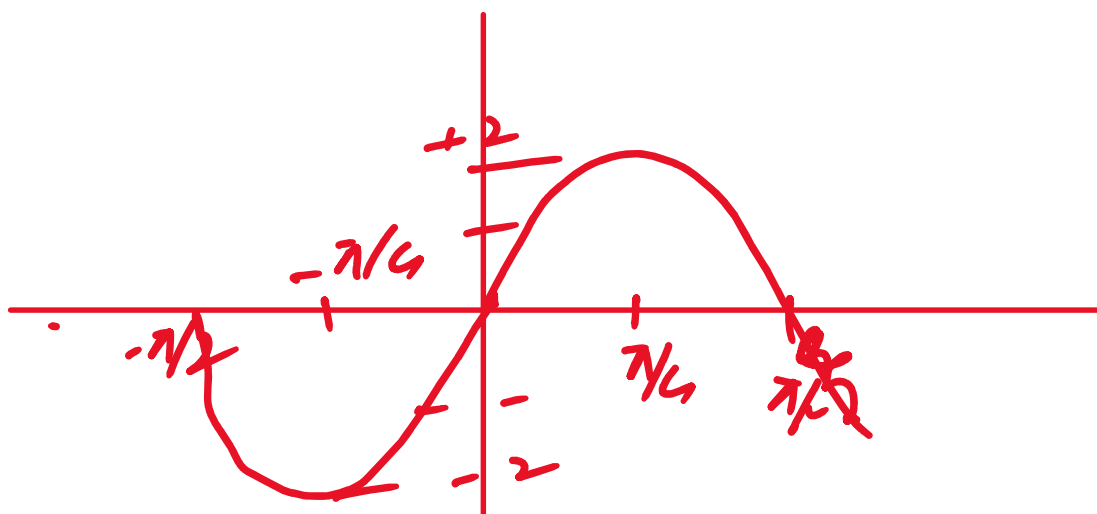
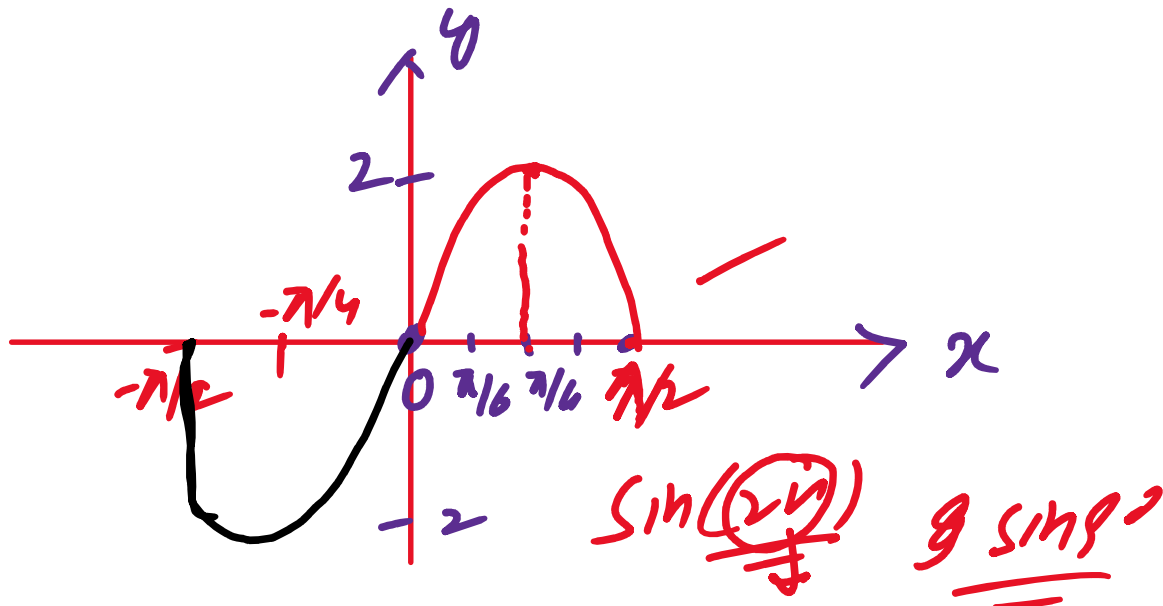
$$2x = 2\pi$$

$$x = \frac{2\pi}{2} = \pi$$

$\pi = \frac{\pi}{3} = \frac{\pi}{2}$

$$\downarrow$$

x	$-\frac{\pi}{2}$	$-\frac{\pi}{3}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\rightarrow$
y	0	$-\sqrt{3}$	-2	0	$\sqrt{3}$	2	0	$\rightarrow$



If x is very small such that its square and higher powers can be neglected, then show that:

$$\frac{(8 + 2x)^{\frac{2}{3}}}{(2 + x)(\sqrt{4 - 2x})} \approx 1 - \frac{x}{12} \quad \checkmark$$

$$\text{L.H.S} = (8 + 2x)^{\frac{2}{3}}$$

$$\boxed{|x| < 1}$$

$$\underline{\underline{(-1, 1)}}$$

$$(2+x)(\sqrt{4-2x})$$

As we know that Binomial Series is $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

We will neglect square and higher terms in expansion of Binomial Series -

$$= \frac{(8+2x)^{\frac{2}{3}}}{(2+x)(\sqrt{4-2x})}$$

$$= \left(8 \left(\frac{8+2x}{8}\right)\right)^{\frac{2}{3}}$$

$$(x \cdot y)^n = x^n \cdot y^n$$

$$(2^3)^{\frac{2}{3}} = 8^{\frac{2}{3}} \left(1 + \frac{x}{4}\right)^{\frac{2}{3}}$$

$$1 + \frac{1}{2}x + \dots$$

$$\frac{2 \left(1 + \frac{x}{2}\right) 4^{\frac{1}{2}} \cdot \left(1 - \frac{x}{2}\right)^{\frac{1}{2}}}{}$$

$$= \frac{4 \left(1 + \frac{2}{3} \cdot \frac{x}{4} + \text{Higher terms ignored}\right)}{}$$

$$4 \left(1 + \frac{x}{2}\right) \left(1 - \frac{x}{4}\right)$$

$$= \frac{\left(1 + \frac{x}{6}\right)}{}$$

$$1 - \frac{x}{4} + \frac{x}{2}$$

$$= \frac{\left(1 + \frac{x}{6}\right)}{}$$

$$\left(1 + \frac{x}{4}\right)$$

$$= \left(1 + \frac{x}{6}\right) \left(1 + \frac{x}{4}\right)^{-1}$$

$$= \left(1 + \frac{x}{6}\right) \left(1 - \frac{x}{4}\right)$$

$$1 - \frac{x}{4} + \frac{x}{6}$$

$$\begin{aligned}
 &= 1 - \frac{x}{4} + \frac{x}{6} \rightarrow \cancel{1} \\
 &\neq 1 - \frac{x}{12} = 1 + \frac{-3x + 2x}{12} \\
 &(1+x)^n \rightarrow 1+nx
 \end{aligned}$$

Use binomial theorem to expand $\sqrt{101}$ and $\sqrt{99}$ separately. Multiply the two expansions and approximate the final answer rounded to three decimal places.

Sol

$$\sqrt{101} = (100+1)^{\frac{1}{2}}$$

$$= \left(100 \left(\frac{100+1}{100}\right)\right)^{\frac{1}{2}}$$

$$= (100)^{\frac{1}{2}} \left(1 + \frac{1}{100}\right)^{\frac{1}{2}}$$

$$\sqrt{101} = 10 \cdot \left(1 + \frac{1}{2} \times \frac{1}{100} + \text{Higher terms repeat}\right)$$

$$\sqrt{101} = 10 \left(1 + \frac{1}{200}\right)$$

$$10 \times 1.005 = 10.05$$

$$\begin{aligned}
 \sqrt{99} &= \left(100 - 1\right)^{\frac{1}{2}} \\
 &= (100)^{\frac{1}{2}} \left(1 - \frac{1}{100}\right)^{\frac{1}{2}}
 \end{aligned}$$

$$\frac{(\frac{1}{2})(\frac{1}{2}-1)}{2} \left(\frac{1}{100}\right)^2$$

$$\begin{array}{r}
 -1.25 \times 10^{-5} \\
 \hline
 0.0000125
 \end{array}$$

$$\sqrt{101} = 10 \times 1.005 = 10.05$$

$$\boxed{\sqrt{101} = 10.050} \rightarrow (1)$$

$$\begin{array}{r} 10.0498 \\ \hline 10.050 \end{array}$$

$$\begin{aligned} \sqrt{99} &= (100 - 1)^{\frac{1}{2}} = \left(100 - \frac{100 \times 1}{100}\right)^{\frac{1}{2}} \\ &= 100^{\frac{1}{2}} \left(1 - \frac{1}{100}\right)^{\frac{1}{2}} \end{aligned}$$

$$\sqrt{99} = 10 \left(1 - \frac{1}{200} + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(-\frac{1}{100}\right)^2}{2}\right)$$

$$\begin{aligned} \sqrt{99} &= 10 \left(1 - \frac{1}{200} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(\frac{1}{10000}\right)}{2}\right) \\ &= 10 \left(1 - \frac{1}{200} - \frac{1}{80,000}\right) \end{aligned}$$

$$\boxed{\sqrt{99} = \underline{\underline{9.950}}} \rightarrow (2)$$

$$\sqrt{101} \times \sqrt{99} = (10.050)(9.950)$$

$$\boxed{\sqrt{101} \times \sqrt{99} = 99.998}$$

A college has 8 Science teachers and 6 Arts teachers.
In how many ways can a committee of 6 teachers be formed if it includes:

- a. Exactly 4 Science teachers and 2 Arts teachers
b. 5 Science teachers and 1 Arts teacher
c. At least 5 Science teachers
d. At most 1 Arts teacher



a) No. of ways = $\binom{8}{4} \binom{6}{2}$

$$= 70 \times 15$$

$$= \underline{1050}$$

b) No. of ways = $\binom{8}{5} \binom{6}{1}$

$$= 56 \times 6$$

$$= \underline{336}$$

A college has 8 Science teachers and 6 Arts teachers.
In how many ways can a committee of 6 teachers be formed if it includes:

A college has 8 Science teachers and 6 Arts teachers. In how many ways can a committee of 6 teachers be formed if it includes:

- Exactly 4 Science teachers and 2 Arts teachers
- 5 Science teachers and 1 Arts teacher
- At least 5 Science teachers
- At most 1 Arts teacher

c) No. of ways = $\binom{8}{5} \binom{6}{1} + \binom{8}{6} \binom{6}{0}$

$$= 56 \times 6 + 28 \times 1$$

$$= 336 + 28$$

$$= 364$$

d) $\binom{6}{0} \binom{8}{6} + \binom{6}{1} \binom{8}{5}$

$$= 28 + 336$$

$$= 364$$

Solve the homogeneous system of equations using Gaussian Elimination method:

$$\begin{aligned} x + 2y - z &= 0, & 2x + 3y - 2z &= 0, \\ -x - y + z &= 0 \end{aligned}$$

Sol $A_b = \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 2 & 3 & -2 & : & 0 \\ -1 & -1 & 1 & : & 0 \end{bmatrix}$

$$R \sim \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 0 & -1 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ R_2 + (-2)R_1 \rightarrow R'_2 \\ R_3 + R_1 \rightarrow R'_3 \end{array}$$

$$R \sim \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 1 & 0 & : & 0 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ (-1)R_2 \rightarrow R'_2 \end{array}$$

$$R \sim \begin{bmatrix} 1 & 2 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix} \quad \begin{array}{l} \text{By} \\ R_3 + (-1)R_2 \rightarrow R'_3 \end{array}$$

$$x + 2y - z = 0 \rightarrow \textcircled{1}$$

$$y = 0 \rightarrow \textcircled{2}$$

$$0z = 0$$

The AR Rank $A = \text{Rank } A_b = 2$
 and less than No. of unknown,
 So, the given system has infinite
 Let $z = t$ $y = 0$

Let $\boxed{z=t}$, $\boxed{y=0}$

(1) $\Rightarrow x + 2(0) - z = 0$
 $\boxed{x=t}$ R

Thus $\boxed{x=t, y=0, z=t}$

Given that $5, (5+3)2, (5+6)2^2, (5+9)2^3, \dots$	
a. Show that given sequence is arithmetic geometric.	3+5
b. Find the sum of first n-terms of the sequence.	

A.P
 a_1, d
G.P
 a_1, r

Sol a) As Arithmetic-geometric
sequence has a form $a_n = \underline{\underline{(a+dn)r^n}}$

Where a = 1st term
 d = Common difference

n = term No

r = Common ratio

1 2 3 4

Given that $5, (5+3)2, (5+6)2^2, (5+9)2^3, \dots$

Given that $5, (5+3)2, (5+6)2^2, (5+9)2^3, \dots$

The terms of given sequence are

$$a_1 = (5+0)2^0 =$$

$$a_2 = (5+3)2^1$$

$$a_3 = (5+6)2^2$$

$$a_4 = (5+9)2^3$$

$$a_1, d=3-0=3$$

$0, 3, 6, 9$

$$a_n = 0 + (n-1)3$$

$a_n = 3(n-1)$

✓

Thus General term

$a_n = (5 + 3(n-1))2^{n-1}$

✓✓

Arithmetic part

$$a_1 = 5$$

$$a_2 = 8$$

$$a_3 = 11$$

$$a_4 = 14$$

Thus A.P. is $5, 8, 11, 14$

In ... diff $d = 3$

$$\boxed{\text{Common diff } d = 3}$$

$$\boxed{a_1 = 5}$$

Geometric seq $2^0, 2^1, 2^2, 2^3, \dots$

$$\text{Common Ratio } r = 2$$

Hence given seq. is Arithmetic-Geo-geometric

$$b) \quad a_n = (5 + 3(n-1)) 2^{n-1}$$

$$a_n = (5 + 3n - 3) 2^{n-1}$$

$$a_n = (3n + 2) 2^{n-1}$$

$$S_n = \sum_{n=1}^n a_n \quad \checkmark$$

$$S_n = \sum_{n=1}^n (3n + 2) 2^{n-1}$$

$$S_n = \sum_{n=1}^{\infty} (5n+3)2^n$$

$$S_n = \frac{a}{1-r} + dr \frac{(1-r^n)}{(1-r)^2} - \frac{(a+nd)r^n}{1-r}$$

$$S_n = \frac{5}{1-2} + (3)(2) \frac{(1-2^n)}{(1-2)^2} - \frac{(5+n(3))2^n}{1-2}$$

→

- The vertices of a tetrahedron are $A(1,2,3)$, $B(2,3,1)$, $C(3,1,2)$ and $D(4,2,1)$
- Find volume of the tetrahedron $ABCD$
 - Using same vectors, find the volume of the parallelepiped formed by $\vec{AB}$, $\vec{AC}$ and $\vec{AD}$.

sol)

$$\vec{AB} = (2-1)\underline{i} + (3-2)\underline{j} + (1-3)\underline{k}$$

$$\vec{AC} = (3-1)\underline{i} + (1-2)\underline{j} + (2-3)\underline{k}$$

$$\vec{AD} = (4-1)\underline{i} + (2-2)\underline{j} + (1-3)\underline{k}$$

$$\vec{AB} = \underline{i} + \underline{j} - 2\underline{k}$$

$$\vec{AC} = 2\underline{i} - \underline{j} - \underline{k}$$

$$\vec{AD} = 3\underline{i} + 0\underline{j} - 2\underline{k}$$

$$= \frac{1}{6} [\vec{AB} \cdot \vec{AC} \times \vec{AD}]$$

$$\text{Volume of Tetrahedron} = \frac{1}{6} \begin{vmatrix} 1 & 1 & -2 \\ 2 & -1 & -1 \\ 3 & 0 & -2 \end{vmatrix}$$

$$\text{Jefrahull} = 6 \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 3 \\ 0 & 0 & 0 \end{vmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 1(2-0) - 1(-4+3) \\ -2(0+3) \end{bmatrix}$$

$$= \frac{1}{6} \{ 2 + 1 - 6 \}$$

$$= \frac{1}{6} [-3] = -\frac{1}{2} \checkmark$$

$$\text{Vol} = |\vec{AB} \cdot \vec{AC} \times \vec{AD}| = \frac{1}{2} \text{ m}^3$$

b)

Vol of parallelepiped = 6x Volume of Tetrahedron

$$\frac{(\vec{x_1}, \vec{x_2})}{d} = 6 \times \frac{1}{2} = 3 \text{ m}^3$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Prove the fundamental law of trigonometry

$$\cos(\alpha - \beta) = \cos\alpha \cos\beta + \sin\alpha \sin\beta$$

where α, β (real angles) are in standard position and $\alpha > \beta$.

08

1.1) Consider a unit circle ($r=1$)

10

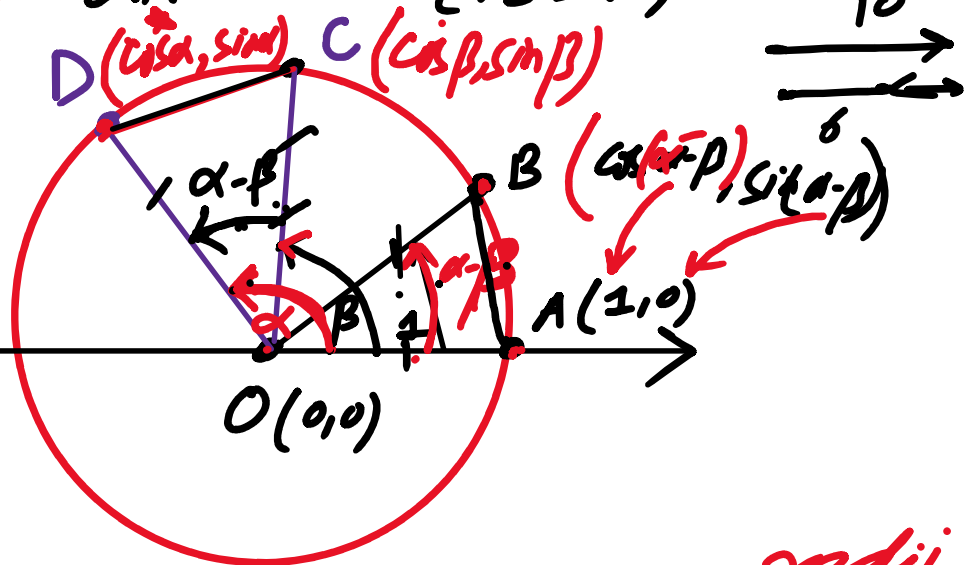
Sol Consider a Unit Circle ($r=1$)

Take $\alpha > \beta$

$\angle AOD = \alpha$

$\angle AOC = \beta$

$\angle COD = \alpha - \beta$



As OA, OB, OC & OD are radii
of unit circle $\Rightarrow OA = OB = OC = OD = 1$

Make point B such that

$$\angle AOB = \alpha - \beta$$

Join C to D

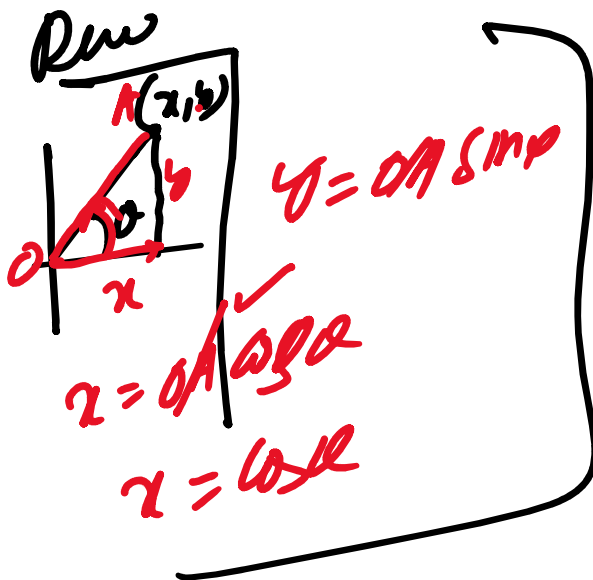
& A to B

As $\triangle COD$ and

$\triangle AOB$ are

Congruent (SAS)

$$\therefore |CD| = |AB|$$



$$\text{So } |\overline{CD}| = |\overline{AB}|$$

$$\Rightarrow |\overline{CD}|^2 = |\overline{AB}|^2$$

$$(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$$

$$= (\cos(\alpha - \beta) - 1)^2 + (\sin(\alpha - \beta) - 0)^2$$

$$(\cos x)^2 = \cos^2 x$$

~~$$\cos^2 x/2$$~~